Project Design Phase
Solution Architecture

Date	2 March, 2026
Team ID	MBA-041
Project Name	Heart Disease Analysis using Tableau
Maximum Marks	4 Marks

Solution Architecture:

The Solution Architecture describes the structural design of the system, outlining how different components interact to deliver the final analytics solution.

3.1 Data Layer

The data layer is responsible for storing and managing the heart disease dataset. The dataset is stored in a MySQL database, which provides a structured environment for managing large volumes of data. This layer ensures data integrity, consistency, and efficient retrieval.

3.2 Data Processing Layer

The processing layer handles data transformation and preparation. SQL queries are used to:

- Filter irrelevant or redundant data
- Aggregate values for analysis
- Create derived metrics
- Structure data for visualization

This layer ensures that the dataset is optimized for analytical use.

3.3 Analytics & Visualization Layer

This is the core layer where Tableau is used to create visualizations, dashboards, and stories. Tableau connects directly to the database and retrieves processed data for analysis. The visualizations include charts, graphs, and comparative visuals that highlight patterns and correlations.

Key features of this layer include:

- Interactive dashboards
- Real-time filtering

- Drill-down analysis
- Visual storytelling

3.4 Application Layer

The application layer enables web-based access to dashboards using Flask. This layer acts as a bridge between the visualization environment and end users, allowing dashboards to be embedded in a web interface for broader accessibility.

3.5 User Layer

The final layer consists of end users such as healthcare professionals, policymakers, and individuals. Users interact with the system through dashboards and stories, applying filters and exploring insights to support decision-making.